

Akshay Pal

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Summary

Software Engineer and Technical Lead experienced in designing, building and owning data-intensive distributed systems using Java, Spring Boot and Kafka. Strong background in backend engineering, including event-driven architectures, scalable APIs, data-processing pipelines, cloud technologies and production troubleshooting. Proven ability to lead engineering delivery, translate complex requirements into reliable software and collaborate effectively with technical and non-technical stakeholders.

Holds active UKSV Security Check clearance.

Technical Skills

Languages: Java 17–25, Python, C++20, SQL, C#

Backend Engineering: Spring Boot 3–4, RESTful APIs, gRPC, JDBC, microservices

Distributed Systems: Kafka, event-driven architecture, asynchronous processing, consumer groups, distributed data pipelines

Data: PostgreSQL, Elasticsearch, Redis, Caffeine

Cloud and Platform: AWS, Kubernetes, Docker, Jenkins, CI/CD

Testing: JUnit, Mockito, Testcontainers

Build and Tooling: Maven, CMake, Git

Practices: TDD, Agile, Scrum, Microservices Architecture

Experience

BAE Systems Digital Intelligence – Senior Software Engineer & Tech Lead

Sept 2022 – present

Career Sabbatical (May 2026 – August 2026)

- Took an approved career sabbatical while remaining employed at BAE Systems Digital Intelligence.

Project 3 (International Government) – Senior Software Engineer (Jan 2026 – May 2026)

- Modernised five independently deployed backend services **from Java 11 and Spring Boot 2 to Java 25 and Spring Boot 4** within a high-throughput network-analytics platform.
- Developed Java services within a distributed network-analytics platform **communicating via gRPC and Kafka** and responsible for **analysing billions of packet observations and tens of millions of derived records daily**.
- Integrated **two new threat indicator data feeds** into distributed ingestion and processing pipelines, expanding the data available for downstream analytics.
- Collaborated with engineers and stakeholders across **three teams in two countries** to coordinate integration changes and minimise disruption to existing platform users.

Project 2 (Home Office): Senior Software Engineer → Technical Lead (April 2023 – Jan 2026)

- Progressed from Graduate Software Engineer to Senior Software Engineer and subsequently Technical Lead within **22 months**, selected based on technical performance and delivery impact.
- Led a team of **five** software engineers, coordinating backend delivery, allocating work, reviewing technical designs and code, and supporting engineers with implementation decisions.
- Owned backend architecture and delivery across **five** independently deployed Java and Spring Boot microservices within a critical national infrastructure platform **processing approximately tens of millions of records per day**.
- Designed and delivered event-driven data-processing pipelines using Kafka and PostgreSQL, supporting **hundreds of thousands of events per second at peak, across twelve partitions and six topics**.

- Designed and implemented **REST API endpoints** connecting internal microservices and integrating the platform with two external Home Office systems, requiring collaboration with external stakeholders and engineers.
- Enabled acquisition, standardisation, aggregation, enrichment and analysis and targeting workflows by defining reliable API contracts and data flows between independently owned systems.
- Developed unit, integration and component tests using JUnit, Mockito and Testcontainers, maintaining a minimum of **95%** coverage across five microservices.
- Used Elasticsearch and Grafana to trace records across distributed services, diagnose production issues and identify processing bottlenecks.
- Coordinated delivery across **five concurrent technical workstreams and three neighbouring engineering scrum teams**.
- Delivered **fortnightly production releases**, deploying the work completed in that sprint.
- Mentored **five** engineers through code reviews, pair programming, technical workshops and structured development objectives.

Project 1 (HMRC): Graduate Software Engineer (Sept 2022 – April 2023)

- Supported an AWS cloud migration, developing scripts to improve reliability of system transitions.
 - Built data validation pipelines using Python and SQL, improving data quality across systems.
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Open-Source Engineering

Bloch – Creator and Lead Engineer

- Designed and created Bloch, a strongly typed hybrid quantum programming language and runtime written predominantly in C++20.
 - Applied object-oriented design and systems-programming techniques across approximately 12,500 lines of first-party production C++ and 4,100 lines of test code, structured into six library targets and one command-line executable.
 - Built a suite of 290+ automated tests covering lexical analysis, AST construction, parsing, semantic validation, runtime behaviour and end-to-end command-line workflows.
 - Created cross-platform build, testing and release automation using CMake and GitHub Actions, producing installable packages with checksums for four operating-system and architecture targets: Linux x64, macOS x64, macOS ARM64 and Windows x64.
 - Authored 350+ commits and managed five version tags, including three packaged public releases, while owning architecture, implementation, testing, documentation and release engineering end to end.
 - Maintained 19 user guides alongside 36 engineering design, architecture and contributor documents.
 - Authored a peer-reviewed software paper documenting Bloch's design and implementation, published by the Journal of Open Source Software in 2026.
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Certifications

[Foundational C# with Microsoft](#) – freeCodeCamp & Microsoft

Education

MSc Quantum Technology Applications and Management – University of Sussex

- Distinction

MEng Engineering Science – University of Oxford

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